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Business process management framework: systematic review of the trends, potentials and future

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ABSTRACT

The dynamics of business processes necessitate effective management to achieve an optimum outcome. In this study, a systematic literature review was conducted to explore the trends, potentials and future of business process management (BPM) and modelling frameworks for effective business performance. Initially, a total number of 5630 articles were obtained from different academic databases, and 83 articles were reviewed due to their relevance to the topic. The review highlights the importance and limitations of the BPM framework, and some outcomes from organisations where it has been implemented. Based on the outcomes of the review, this study proposes a BPM framework comprising of the basic components that organisations can employ to identify, assess, and manage risks proactively. It further noted the positive influence of BPM on some organisations although some reported cases of under-implementation due to the dynamics and complexity of the business environment and misalignment with management priorities. Emerging trends indicate the future of BPM frameworks will move from a static process modelling tool to a more sophisticated data-driven and adaptive tool integrating digital technologies such as digital twin. The outcome of this study can assist business experts and organisations understand business workflows for effective analysis and improvement.

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1. Introduction

The ever-changing complexity of the business environment demands that business processes be managed effectively to achieve an optimum outcome. The competitive nature of the present business environment, also demands that organisations model the business processes to gain proper understanding for effective workflow, information and business management, and improved performance. This will also assist organisations to achieve their bottom-line of profitability through proper definition of business processes, re-engineering, control, and implementation of continuous improvement strategies (Harmon, 2020). Business Process Management (BPM) is concerned with the use of various methods for the identification, analysis, modelling, and improvement, of business processes while business process modelling is a subset of the BPM which is a common approach for obtaining and representing business information for effective analysis, communication and decision making. The business process modelling is a management technique for the visualization of business processes to promote effective monitoring (Harmon, 2020). The business process modelling shows the state of a business, the tasks allocated, the sequence of task implementation, resources deployed as well as the personnel responsible and the assigned roles. Process models describe the operations and activities of an organisation to effectively mitigate the risks and allow for process optimization. Thus, the progress of organisations in terms of business performance can be visualised, communicated and documented using models (Association of Business Process Management Professionals, 2020). Business process modelling involves the representation or creation of current or future business process and activities via a model (Association of Business Process Management Professionals, 2020). Management strategies such as TQM, Business Process Improvement (BPI), and BPR

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help organisations gain competitive although there are limitations such as limited information and insufficient analysis that led to the concept of business process modelling concept (Ivančić, 2019). The concept is now gradually emerging and embraced as a comprehensive approach to analysing business for performance and process improvement (Cardoso & Aalst, 2009).

It is an important aspect of process improvement or re-engineering by which process models are produced manually or in a digital form using software (Harmon, 2020). Business process modelling can provide a graphical or visual representation of a piece or chains of processes (including major or supporting) (Association of Business Process Management Professionals, 2020). It finds extensive application in the business sector and other related fields such as quality management (Ivančić, 2019). It starts with the identification and mapping of the current state or business process known as the 'AS-IS' process to the estimated or projected or optimised process known as the 'TO-BE' process. The optimisation is usually conducted using process informatisation, digitisation or process steps change (Ivančić, 2019). The outcome of the expected process (AS-IS) and the prediction of its performance will be carried out against the selected key performance indices (KPIs). Standard modelling notation is usually used to denote how some activities must be carried out in a definite order highlighting all the instruments required to execute the task. By this, individuals who understand modelling notation can easily comprehend the process models and the activities represented in a standard language (Cardoso & Aalst, 2009). However, standard languages, may not be fully comprehensive although they reflect a business process but standard may lack some modelling characteristics of a specific business domain. Hence, where all the all features of a particular application process models are not sufficiently covered, it might be difficult to understand (Cardoso & Aalst, 2009). The process model also finds application in re-engineering such as descriptive enactment involving process model design to supervise a task and ensure it is implemented automatically (Cardoso & Aalst, 2009).

A review of the business process modelling approaches will provide a valuable insight into its potentials and limitations and how they could be solved. Dumas et al. (2018) underscore the significance of the business process models throughout a business lifecycle. Its understanding as well as the knowledge of the process to modelled are necessary to initiate a robust and representative process model.

Processes can be understood through a combination of observation, theoretical and knowledge sharing with the people who are directly engaged with it on a regular basis (Dumas et al., 2018). This is the very first step to gaining a proper understanding of business processes including the challenges and the possible mitigating strategies, before a process analysis, modelling, reengineering, or automation can be conducted (Dumas et al., 2018).

The aim of this study is to conduct a systematic literature review to explore the potentials of business process modelling for effective business performance and to provide some practical guidelines for the implementation of BPM.

The research questions (RQs) underlying this study are:

1. **RQ1:** What are the existing theories, frameworks and approaches used in BPM?
2. **RQ2:** What are the trends, benefits, and limitations of BPM?
3. **RQ3:** What are the potential applications of BPM and the gaps existing in its framework?

The increasing complexity of business processes amidst a competitive market necessitate the quest for business to be driven to obtain optimal results. To explore the potentials and benefits of BPM, a systematic literature survey was conducted in this study. The literature review highlights the usefulness of the business-modelling framework, its notation, outcome of its implementation by various organisations as well as practical guidelines for its implementation. Based on the outcomes of the literature review, this study proposes a BPM framework comprising of the basic components that organisations can employ to identify, assess, and manage risks proactively.

The study is significant in that it can assist business experts and organisations to understand business workflows, for effective analysis automation, decision-making and improvement. The findings presented in this work can also assist business analysts and organisations to understand improved ways to optimise workflows, analyse and automate current business processes.

The study contributes conceptually and theoretical to the existing literature on BPM especially the current and projected future trends as well as potentials and application.

This study is structured as follows: The first section describes the background to the study, research problem, research aim, significance and contribution to knowledge while [Section 2](#) covers the methodology employed. [Section 3](#) reviewed the literature systematically while [Section 4](#) applied the BPM framework for risk mitigation. [Section 5](#) presents the results and discussions while the last section provides the conclusion, recommendations and direction for future studies.

2. Methodology

This study uses a systematic literature review conducted to explore the capabilities and benefits of BPM. Amongst others, the review highlights the importance of the BPM framework, and some outcomes from various organisations where it has been implemented. Appropriate articles were acquired using the search engines and content analysis of the acquired articles was carried out. The systematic literature review, follows the process of acquisition of articles from academic database and other repository using the keywords search and the application of the selection criteria (Daniyan et al., 2021).

2.1. Research design

This study implements the SLR using the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020). The SLR method ensures a structured, reproducible, and transparent synthesis of existing studies on BPM thus reducing bias and increasing the reliability of the findings.

2.2. Search strategy

Searches were conducted across different sources including Science Direct, IEEE, ACM Digital library, Scopus library, Google scholar and Web of Science. The search terms combined keywords and Boolean operators such as: ('Business Process' OR 'Business Workflows' OR 'Business Process Management' OR 'Business Process Modelling' OR 'BPM' OR 'Process Modelling' OR 'Business Modelling' OR 'Process Notation').

AND

('framework' OR 'model' 'methodology' OR 'approach' OR 'automation').

Initially, a total number of 5630 articles were obtained from different academic database but only 83 articles were reviewed after the implementation of the selection criteria. The SLR provides a comprehensive overview of evidence on the subject matter (BPM) and assist in identifying the research gaps.

2.3. Inclusion and exclusion criteria

The inclusion criteria were considered: studies proposing or applying BPM frameworks, methods, or notations, empirical, conceptual, or comparative studies relevant to BPM and papers written in English while the exclusion criteria include studies that did not focus on BPM, as well as duplicates or incomplete records.

To ensure transparency, the inclusion and exclusion criteria were explicitly implemented, documented and consistently applied by the authors and the areas of conflict were resolved via some brainstorming sessions. Furthermore, three reviewers' opinions were sought and reconciled with the selection and synthesis of the authors.

2.4. Study selection process

The selection followed PRISMA's four stages namely: identification (This includes the records retrieved from databases using search strings), screening: (Removal of duplicates and irrelevant title or/abstract,

eligibility (full-text review vis-à-vis the selection criteria) and inclusion involving the listing of the final set of studies included for literature synthesis.

2.5. Data extraction and coding

To ensure reproducibility, the data extraction process was structured to ensure information include the bibliographic information (author, year, and source), type of BPM framework or notation, application context (domain, case studies, tools, reported benefits, potentials, limitations, and outcomes as well as research gaps.

To reduce bias and subjectivity, the authors conducted the selection process independently then reviewed the conflicting papers via a brainstorming session by strictly considering the inclusion and exclusion criteria.

2.6. Data analysis and synthesis

The thematic approach was used to analyse the extracted data. BPM frameworks were classified based on modelling approaches, supported tools, and application domains. Comparative analysis was conducted to discuss the areas of intersection, divergences, and limitations across studies.

Process architecture was classified according to organisation processes while business architecture and process management were grouped into three classes namely enterprise architecture (which evaluates the organisation's operations from the technological perspective) and business architecture (this links the organisation's strategy to its capabilities and business operations and processes).

The benefits of the BPM were also categorised as communication, learning and long-term benefits.

For ease of reproducibility, the full documentation of the search strings and selection criteria, was ensured. In terms of transparency, selective reporting was avoided by implementing the PRISMA protocols consistently and rigorously with explicit synthesis and reporting of the studies selected. Protocols were followed consistently to avoid selective reporting.

To ensure rigour and strengthen the SLR methodology, specific measures such as multi-database search, implementation of well-defined inclusion and exclusion criteria, and quality appraisal of studies based on methodological soundness and contribution to BPM research were ensured.

It was ensured that the following checklists were adhered to as part of quality assessment of the selected articles:

- i. Clarity of objectives – Does the study clearly state its aim or objectives and research questions?
- ii. Relevance – Is the study directly related to business process modelling, BPM or business process frameworks?
- iii. Methodological rigour – Is the methodology clearly and sufficiently described and justified (for instance, conceptual, empirical and case study or modelling approach)?
- iv. Evidence and validation – Are the results obtained supported by empirical data, experiments, or case studies?
- v. Transparency of reporting – Are limitations, assumptions, and context clearly described? Are the discussions based on the findings?
- vi. Contribution – Does the study provide novel contributions or insights or practical or theoretical contributions to BPM?

These measures ensure that the proposed BPM framework is based on a comprehensive, reproducible, and transparent synthesis of the existing literature. To ensure transparency, this study adopts the PRISMA 2020 guidelines consisting of a multi-database search strategy where the search strings and eligibility criteria are explicitly defined. Furthermore, the various BPM methods, theories and frameworks, lifecycle etc. were identified and discussed to ensure strengthen rigour of the review.

Figure 1 presents the PRISMA framework employed including the selection criteria for the articles reviewed. It displays the procedural steps undertaken for the selection of the articles reviewed from the

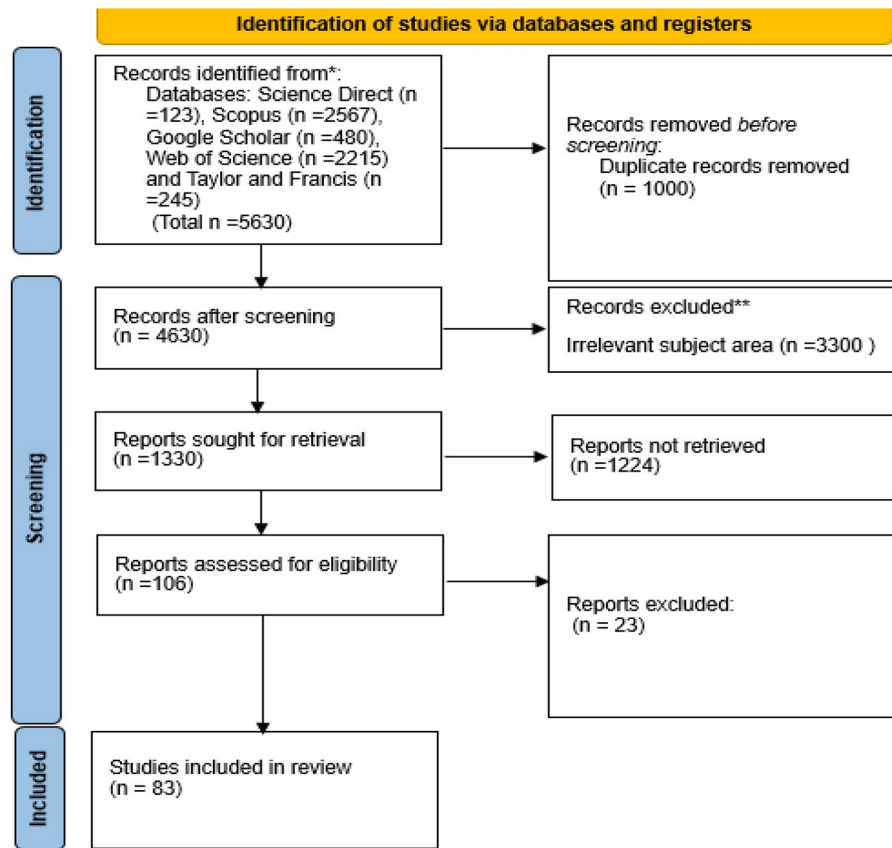


Figure 1. The PRISMA framework employed.
Source: Authors'.

acquisition to their respective repositories followed by the screening via the implementation of the exclusion criteria and the matching of the inclusion criteria with the selected articles.

Table 1 shows the breakdown of the articles according to the year of publication. To enhance rigour, search strings were adapted for each database, and the review period was restricted to 2000–2025 covering 26 years to reflect the evolution of BPM methods over the digital age. However, four (4) few foundational textbooks that fall outside this review period were included because of their theoretical robustness. Only peer-reviewed articles with few grey literatures were considered which comprise of empirical, conceptual, theoretical and review studies.

3. Review of literature on business process modelling

Business process modelling is a vital part of information system design, business engineering or re-engineering (Jeston & Nelis, 2006). It summarises and externalises the manner in which business activities are conducted supported by an information system (Jeston & Nelis, 2006). Business process models of organisations may be different depending on the business process, business analysts, and model designers and the involvement of other people. The nature support provided by the vendors of business process modeling tools to the end users as well as the educational services provided by academics determine the nature of business process models and its success (Indulska et al., 2009).

The choice of the modelling language used to represent business activities is also an important consideration. Business process modelling languages standardise the representation, presentation and communication to the different stakeholders (Jeston & Nelis, 2006). The analysis of the uses of the process models largely determines the choice of the language that will be employed. If the process models are regularly shared with other stakeholders, a standard business process modelling language may be more suitable (Cardoso & Aalst, 2009). Business process modelling is interesting in that it comprises of vast

Table 1. Breakdown of the articles according to the year of publication.

S/N	Year	Number of articles
1.	1951	1
2.	1968	1
3.	1972	1
4.	1977	1
5.	2002	1
6.	2003	1
7.	2005	2
8.	2006	1
9.	2008	1
10.	2009	6
11.	2010	4
12.	2011	1
13.	2012	1
14.	2013	1
15.	2014	3
16.	2015	5
17.	2016	2
18.	2017	4
19.	2018	12
20.	2019	5
21.	2020	5
22.	2021	5
23.	2022	3
24.	2023	6
25.	2024	7
26.	2025	3
Total		83

number of different techniques that can be used to represent a business process and the individual techniques are unique based on their requirements (Tangkawarow & Waworuntu, 2016). Thus, process models can promote effective communication needed for process improvement and can also help to visualise the vital stages in a process, as well as the interrelationships or interdependencies among the process models and how they are related to each other. It can also show the people and systems involved in the implementation of the various steps, as well as at the communications that takes place with the customers and other external parties (Delavari et al., 2010). Thus, process model is critical to process improvement (APQC, 2020) as important information can be extracted from the process models (Delavari et al., 2010).

Figure 2 presents the Dalberg process modelling framework for BPM and improvement. It is a practical-oriented framework for business process analysis developed for mapping, diagnosis and for improving complex processes. Contrary to the BPM frameworks, the Dalberg process modelling framework underscores problem-driven analysis, and outcome-focused process redesign with strategic alignment to the stakeholders. The figure depicts that business organization can gain an increasing value when process models are practiced in the right context. The framework models process end-to-end, capturing both formal workflows and informal practices, as well as major decision points, and organisation's challenges. It also adds multi-stakeholder and institutional context to business processes to capture and align inputs from the stakeholders and the organization. The framework divides processes into inputs, activities, decision gates, actors, outputs, and outcomes. This allows effective analysis to determine the point where wastes are generated or where there is loss of value, delay or misalignment. The framework connects process performance at every stage to the strategic objectives and deliverables or outcomes, such as effective service delivery and sustainability. Hence, it can be deployed as a diagnostic and decision-support tool, to assist helping organisation's test policy interventions, and rank reform effectiveness. Although it is less formal compared to the BPM because it integrates informal practices but it can complement the traditional BPM by capturing real-world business scenarios rather than the ideal process flows.

Thus, a process modelling framework can guide decisions such as resource allocation, modelling tools and techniques, responsibilities of the stakeholders, and other factors in order to maximise the value added to an organisation (APQC, 2020).

The survey conducted by Harmon (2020) shows that 72% to 81% respondents employ process models for clarification, communication, process redesign or improvement while 51% to 57% of the respondents

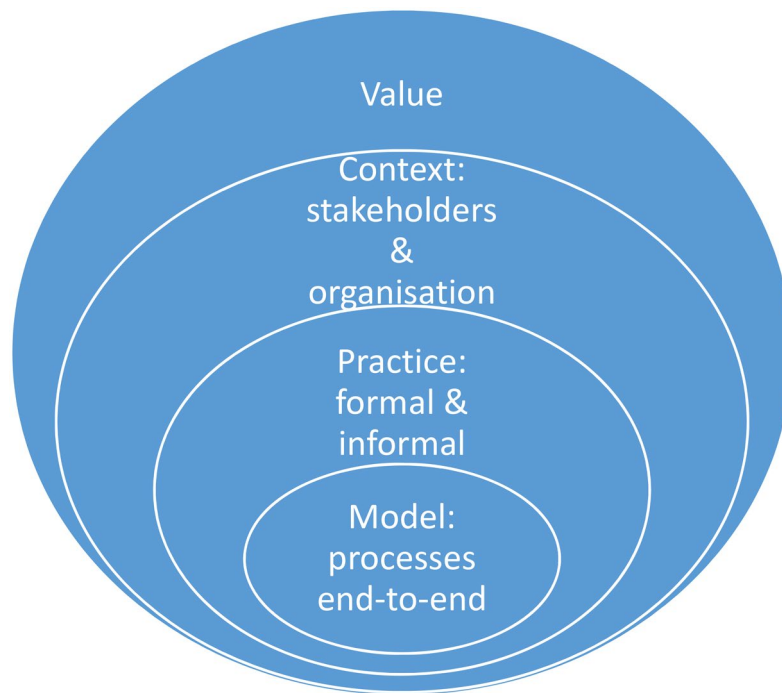


Figure 2. The Dalberg process modelling framework (APQC, 2020).

use process models for transformation initiative, process documentation, and as a preparation for software development (Harmon, 2020). 27% of the respondents used it in combination with Enterprise resource planning (ERP) software project or as a preparation for initiating a business process management system (BPMS) (Harmon, 2020).

Harmon (2020) also found that 57% of the process experts use generic approaches to conduct process modelling while 42% use standardized approaches such as the lean or six sigma or lean sigma, while 15% of the respondents use a different approach. Some combines generic and established approaches.

Business process modelling is an important tool for process representation and analyses. The maps allow easy visualisation of the connections between systems, which assist in aids in the identifying process gaps in order to determine the need for automation or not (Xu et al., 2009). Reijers (2021), stressed need to record and define business processes in an effective manner as this necessary for building a system, and to establish the performance measurement and monitoring. The functional approach to business process modelling is also significant factor to consider as this describes the processes and their structures and defines the goal that must be achieved. Each process is broken down into smaller units known sub-processes (Cardoso & Aalst, 2009).

Cardoso and Aalst (2009) state that processes may be elementary or composite (Cardoso & Aalst, 2009). Elementary processes are processes that cannot be decomposed while composite processes, can be broken down into the constituents' *sub-processes*. A *sub-process* refers to a role name that denotes the different processes (Cardoso & Aalst, 2009) while the behavioural perspective defines the dependencies between processes and describes how the processes must be executed in a definite order. Furthermore, the goal of each process, tasks, and management are defined in the behavioural perspective (Loucopoulos & Heidari, 2014). In addition, the organisational perspective defines the people and the assigned roles for the execution of a defined process are defined (Cardoso & Aalst, 2009).

3.1. Business process modelling notation (BPMN)

The objective of BPMN is to help business organisations comprehend their internal processes (Flowers & Edeki, 2013). The development of models for the representation of processes becomes imperative for large organizations several interconnected processes (Flowers & Edeki, 2013). The visualization of the processes via a process model is essential because of the changes in processes due to

re-engineering, introduction of new applications, etc. Also process models can assist new employees and the team to comprehend the process functionality and other related activities. Thus, it can serve as a roadmap that can assist employees to visualize the different operations of the sections and how their activities fit into the overall activities of the organisation (Flowers & Edeki, 2013). Thus, to understand the various business process modelling standards, organisations can classify business processes into different groups with similar characteristics and functions (Ko et al., 2009). The business process modelling standards address at least one of four phases of its lifecycle. BPMN can ensure data compactness, and compactness of business process designs across different standards. According to Flowers and Edeki (2013), BPMN has grown into maturity as a standard for creation of process models. This implies that the interoperability integrates both the technological and ergonomic factors. Organisations can leverage on the merits of ease of development to document and disseminate their operations effectively (Flowers & Edeki, 2013). Flowers and Edeki (2013) further stated that BPMN may not instantly enable people who are unable to articulate their processes or assist people who cannot comprehend business activities or processes without proper documentation, or knowledge base (Flowers & Edeki, 2013). However, it is a useful technique for specific modelling application and representation especially when there is a clear process map or proper documentation of the business processes (Ko et al., 2009).

3.2. Process architecture

The goal of the process architecture to provide a visual illustration of business processes (Dumas et al., 2018). There is a need to consider the complexity and interrelationships of the organisation while defining its process architecture. To address an organisation's complexity, there is a need to first differentiate the different types of processes (Dumas et al., 2018) such as primary processes, support processes and management processes.

Primary processes are end-to-end processes (usually cross-functional) which adds direct values to the customers. They are usually termed 'core' processes that are reflective of the activities that an organisation must perform to attain set goals with optimum customers satisfaction. The value chain comprises of these processes which each adds value to the preceding ones as determined by the rate of production or the quality of product or service delivered which translate to customer value (Association of Business Process Management Professionals, 2020). The primary processes refer to the creation of organization's core value, in terms of delivery of quality products and services that consumers pay for.

Support processes are designed to aid the primary processes through effective resource and infra-structural management. These include information communication technology (ICT) facilities, finance, effective human resource management, indirect procurement, hardware sourcing, furniture, and stationery, accounting, financial management, and legal services etc. The major difference between the primary and support processes is that primary process provides direct value to the customers while support processes do not. Support processes are usually linked to some specific areas in the organisation and enable them to function effectively through effective management, resource mobilisation and proper scheduling amongst others (Dumas et al., 2018).

Management processes are aimed at measuring, monitoring, and controlling business activities. It is considered social in nature due to the fact that it comprises of a series of activities that lead to the realization of organization's set goals objectives (Newmann et al., 1972). Jesus et al. (2015), define management processes as procedures that relates to the control of the internal activities of an organisation. Furthermore, it entails the processes that aid the translation of strategic requests into BPM services (Jesus et al., 2015).

It ensures the formulation and implementation of primary and support processes in such a way that they meet the functional, operational, financial, regulatory, and legal objectives (Association of Business Process Management Professionals, 2020). Like the support processes, management processes, do not also add direct value to customers, but are imperative to ensure the smooth running of an organisation in line with the required standards and regulations for improved performance (Association of Business Process Management Professionals, 2020).

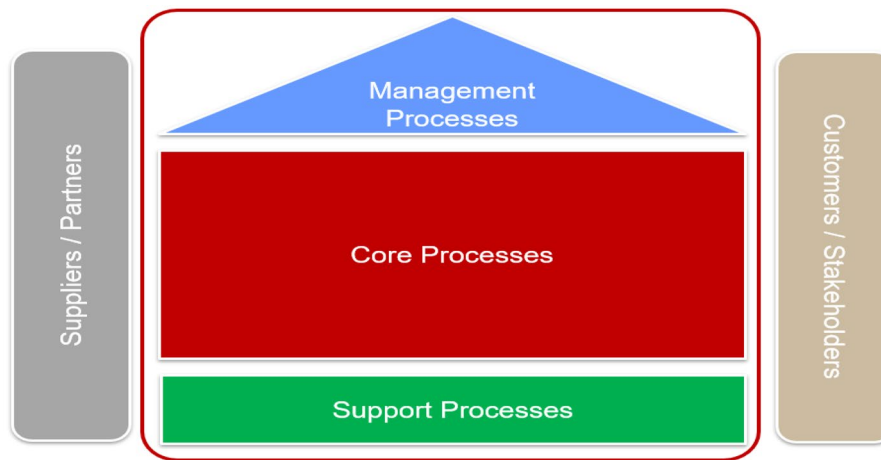


Figure 3. Michael Porter value chain for an organisation's business processes (Dumas et al., 2018).

Another significance of the management processes is that they provide guidance, rules, and standards for the primary and support processes (Association of Business Process Management Professionals, 2020). These include planning, budget and finance regulatory compliance and risk mitigation.

Figure 3 presents the Michael Porter value chain pyramid for an organisation's business processes. It is a strategic framework that can be employed to analyse an organization's business processes by breaking the processes into a sequence of value-adding activities for customers. The main concept is that competitive advantage stems from the effectiveness of the interrelated activities rather than from the organization as a whole. As shown in the Figure, the core processes which contribute directly to the development of products and services may include logistics, operations, marketing and sales, and service (after-sales support and customer service) while support activities that enable the core processes may include infrastructure, governance, finance, planning, human resource management, technology development and procurement. Thus, the Michael Porter value chain is a high-level process architecture that can assist organisations to identify units where values are created, where costs are incurred, or areas where process improvements can produce significant outcomes. Furthermore, it is useful in aligning operational processes with strategy, to enable managers analyse process performance, and detect sources of competitive advantage.

In the Michael Porter's pyramid, the management processes top the hierarchy followed core and support processes. This pyramid can assist organisations to establish effective BPM.

The architecture of BPM can be grouped into three high-level classes (Association of Business Process Management Professionals, 2020). These are:

- Enterprise architecture: this evaluates the organisation's operations from the technological perspective.
- Business architecture: this links the organisation's strategy to its capabilities and business processes (Association of Business Process Management Professionals, 2020).
- Process management: this relates to the end-to-end management of business activities throughout the whole process, including the business lower level and workflow (Association of Business Process Management Professionals, 2020).

Business architecture provides a holistic representation of how ICT and other supporting infrastructure aid process activities as well as the requirements for the delivery of high quality products or services (Association of Business Process Management Professionals, 2020). Thus, business architecture defines business effectiveness. Since organizations battle with many constraints such as increase in competition, dynamic and expanding markets, environment changes increase in customer requirements (Millet et al., 2009; Havey, 2005), organisations must strive to meet these demands. To do so, they must strive to respond effectively to customer's demand and market forces (Xu et al., 2008) through a system of process management. Process management defines how business activities are conducted and how work

changes (Association of Business Process Management Professionals, 2020). Although it may be challenging to achieve process management due to the different types of products, however, information can be harnessed to gain a proper and holistic insight about a business process (Association of Business Process Management Professionals, 2020).

Process management of business processes is effectively used in large organisations as it promotes competitiveness (Delavari et al., 2010). It is a simple management process that seeks to synchronise organisation's efforts and resource to ensure effective performance (Delavari et al., 2010). According to Harmon (Harmon, 2020), the organisation's management need information about the value chains, processes, or sub-processes as well as the measurement and management of specific processes. This is to assist management keep track of variations in performance in order to make informed decision to achieve set goals (Harmon, 2020). Process management also assist organisations to develop the right strategies to implement the business plans. The information also encompasses the interaction among processes, as well as process performance amongst others (Harmon, 2020). Some organisations are developing business tools that can offer the model for making informed decisions (Harmon, 2020).

Neufeld (2020) could not ascertain the difference between business process modelling and other process improvement approaches such as TQM, BPI and BPR and as a result the effect of strategies on business process and organisation's structure could not be substantiated. Despite the fact that business process modelling has evolved over the years, many organisations still rely on existing approaches. Sometimes, organisations may find it challenging to respond to changes or incorporate emerging approach into their business models. Few studies have attempted to establish a relationship between business process modelling and other process improvement approaches. Due to this, organization management may not have a knowledge of the merits of BPM. Thus, to fully implement business process modelling and harness its potentials, there is a need to conduct more research (Reijers, 2021).

3.3. Business process management theory and frameworks

Dumas et al. (2018) define BPM as the approach of work management in an organisation to ensure a significant and positive result. The South African Bureau of Standards (2012) defines a process as inter-related activities that comprises of inputs which produces an output. Sujova and Marcinekova (South African Bureau of Standards, 2012) also define the term 'process management' as a logical way to visualise, measure, and improve business processes for continuous improvement. Sujova and Marcinekova (2015) further clarified that the use of process management principles and methods can assist organisations achieve higher values of Return on Equity.

Many organisations perceive BPM in different business contexts, however, the implementation of process management techniques can promote process improvement in an organisation. It encompasses the development of new, and innovative methods that address the customer needs.

There is a consensus among the literature that process models can be used for process evaluation and improvement (Vom Brocke et al., 2016; Leitz et al., 2018; Mathiesen et al., 2011; Smart et al., 2009; Rosemann & Vom Brocke, 2015). Business process modelling also extends to servicing the needs of the customers (Sujova & Marcinekova, 2015).

BPM is rooted in multiple theoretical frameworks. Core theories underpinning BPM are as follows: System's theory, organisation's theory and Socio-technical System's theory.

3.3.1. System's theory

The system's theory considers an organisation as an interdependent system where processes connect the inputs, to the transformations, then to the outputs. This theory perceives an organisation as an open system comprising of various interrelated and interdependent components that relates with their environments to transform inputs into outputs (von Bertalanffy, 1968). Thus, an organization is considered and analysed as a holistic entity with dynamic interactions and feedback loops amongst the subsystems rather than being analysing isolated tasks or units individually. In the context of BPM, the system's systems theory considers the business process as an integrated and systemic flow of activities across the various units. Resources and information flow are integrated across all the units to create value and any

deviation or in one process can alter the system's balance or affect the overall performance. This highlights the importance of end-to-end process rather than optimization of individual units or sub-systems (Weske, 2024).

The system's theory finds application in the design of BPM process architecture where processes are designed in layers across the strategic, core and operational levels and performance evaluation is done via the KPIs and control loops.

This theory supports BPM as a coordinative and integrative management tool to promote focused on alignment, adaptability, and efficiency.

3.3.2. Organisation's theory

There is also organization's theory that advocates alignment of strategy, structure, and processes to achieve improved performance and overall organisation's goals. Furthermore, a culture of continuous improvement encompassing TQM, Lean, and Six Sigma, have also shaped BPM by enabling improved process performance and reduction in the rate of defect. It accesses how organisation's structures such as governance, stakeholders and institutional contexts as well as formal and informal culture determine process designs, behaviour, implementation and performance (Donaldson, 2021). In the BPM context, the organisation's theory provides relevant insights into process ownership and accountability, coordination across hierarchical and functional boundaries, resistance to process change and institutional constraints such as regulations, compliance and culture. This theory advocates that rather than applying BPM uniformly across different organisations, organisation's contexts should be studied with the BPM deployed accordingly. It stresses that there is no single process that fits all organisations but processes can be designed to fit the various organisation's needs or peculiarities such as size, environment, and technological maturity etc (Scott & Davis, 2000). Hence, this theory provides useful insights as to why BPM standardization across all sectors may succeed in some contexts but fail in others.

This theory is significant to BPM in that it designs processes as workflows with technical and organisation's considerations.

3.3.3. Socio-technical system's theory

The socio-technical system's theory highlights and balances the roles of technology and its enablers as well as human and organizational factors in BPM (Dumas et al., 2018; Newmann et al., 1972; Rosemann & Vom Brocke, 2015). It is an extension of the systems and organizational theories by integrating social factors into technical subsystems (Trist & Bamforth, 1951). It argues that organization's performance and sustainability is linked to the development of systems that aligns technology, processes, and human factors. It advocates for the integration of social and technical subsystems within an organization stressing that an organisation's performance and sustainability is a function of strategic alignment among people, structures, culture, and technology, rather than just technology or processes. In the BPM context, this theory is significant in that it integrates the human and social factors such as human participation, communication, and learning into an organisation's workflows and information technology (IT) systems for effective decision-making. It designs processes tailored to human-system collaboration such that human work is supported by enabling technologies such as automation, analytics, and digital tools rather than constraining it. It considers human-centered process design to prioritize efficiency with job satisfaction, accountability, and adaptability. This is because the traditional BPM often fail for its inability to consider human factor leading to social resistance. This theory reinforces BPM via the development of processes that are socially acceptable, sustainable and technically efficient and finds application in the areas of process automation and digitalization, process mining and decision-support systems, as well as Internet of Things (IoT)-enabled processes and digital twins. Contemporary BPM studies caution against the development of BPM processes or systems that are absolutely technology-dependent without consideration for human factors such as skills and learning. These systems often witness social resistance and low adoption, workarounds, or resistance, even when technically sound. Emerging research adopts the socio-technical theory integrating emerging technologies such as IoT, real-time analytics and digital twins and human decision-making processes into organisation's workflows to deliver a sustained value (Bostrom & Heinen, 1977; Fornari et al., 2025).

3.3.4. Integrated theory for BPM

The combination of these three theories namely the system's theory, organisation's theory and the socio-technical theories can provide a sound theoretical basis for BPM implementation. While the system's theory explains BPM as an integrated and adaptive system of processes, the organization's theory explains how organisation's factors such as structure, and context can influence process design and outcomes. Lastly, the socio-technical systems theory advocates for the consideration of human factors in process workflows for leading to an integrated system of technology, processes, and people.

This integrated theoretical framework can support the digital transformation, sustainability, and continuous improvement of the BPM technique.

The BPM theories stress the strategic importance of processes as organizational assets, while BPM frameworks provide a structured approach for the identification and analysis of processes in order to improve, and sustain process performance. The increasing roles of technology is reflected in the gradual transition of the traditional BPM lifecycle models to intelligent and agile thereby contributing to business process excellence.

Thus, BPM theories integrates management science, organisational theory, and information systems promote continuous improvement of business processes. They indicate that BPM is not just a simply a technical solution but a holistic management approach that integrate and balances the roles of people, culture, governance, and technology.

3.4. Business process management models and applications

A model explains the organisational capability that are important to BPM to assist decision makers understand the necessary actions that organisations must undertake while conducting process management (Rosemann & Vom Brocke, 2015; Liptak & Liptak, 2003).

The six core elements of BPM are listed as follows (Rosemann & Vom Brocke, 2015; Liptak & Liptak, 2003).

1. **Strategic Alignment:** BPM contributes to an organisation's strategic goals. It assesses and processes management initiatives, depending on how they fit into the overall organisation's strategy.
2. **Governance:** Governance deals with the framework for organisation's management and decision making. Thus, BPM should be integrated into the organisational structure.
3. **Methods:** The deployment of suitable methods for business process design, execution, monitoring and control are necessary for the success of BPM.
4. **Information Technology:** Suitable technologies are also needed to drive and automate business process design, execution, monitoring and control.
5. **People:** The success of BPM is also dependent the capacity and expertise of the employees especially when matched with the appropriate business processes.
6. **Culture:** If a tradition of improvement, excellence and innovation must be sustained, then BPM must be integrated into the organisation's value and system.

These six core elements of the BPM are relevant to process management as it identifies the areas where process improvements are needed (Liptak & Liptak, 2003; Vom Brocke & Mendling, 2017; Zur Muehlen & Ho, 2005). Alves et al. (2018) stated that people and culture are intricate factors that should be considered when implementing BPM, while the roles of training in human capacity development is also important. Good organisational culture is an important factor that contributes to organisational behaviour and hence, the success of BPM initiatives (Sujova & Marcinekova, 2015; Schein, 2010).

The application of BPM to Systems Applications and Products (SAP) ensured transparency in processes and execution of key roles, and responsibilities (Reisert et al., 2018). It also helped to mitigate complexities and negative influences on the business and customers (Reisert et al., 2018). Thus, BPM can be applied to improve the performance of individual or sequence of processes. However, its successful application has been reported in limited fields coupled with the fact that there is insufficient in respect of its six core elements.

Table 2. Application of BPM for improvement.

S/N	Problem	Method	Outcome	Reference
1.	Lack of alignment of organisation's plans, with strategic goals, lack of operational processes for monitoring organisations plans coupled with resistance to changes by the employees	Application of the six core elements of the BPM for strategic alignment of organisation's plans and goals and to address the people and culture as key factors causing resistance to changes.	Strategies that promote cultural values and supports the BPM implementation were adopted leading to the elimination of the BPM barriers.	Alves et al. (2018)
2.	Poor incidence management in a telecommunication company leading to customer dissatisfaction.	Analysis of the prevailing scenarios and establishment of new organisational units that implement the core elements, methods and techniques of the BPM.	There were significant improvements in the way incidences are managed thereby resulting in the reduction of average lead time from 13 days to 3.6 days.	Czarnecki (2017)
3.	Waste management	Adoption of the BPM core elements and the deployment of ERP technique to waste management.	There was a transition from waste collection and delivery at designated sites to waste separation, collection, and processing thereby contributing zero-waste through effective waste management	Kovacic et al. (2017)
4	Business complexity and execution of task	The application of BPM to Systems Applications and Products (SAP)	Improved transparency in processes and execution of key roles, and responsibilities. It also helped to mitigate complexities and negative influences on the business and customers.	Reisert et al. (2018)

Table 2 presents some evidences drawn from the literature on the successful implementation of the BPM. The outcome of BPM presented in the Table shows that BPM implementation has contributed significantly to process efficiency across various sectors such as waste management, business organisations and telecommunications.

3.5. Benefits of process management

Process management, can help an organization develop a high-performing processes, which is cost and time effective with high accuracy, flexibility and automation (Rosemann, 2015). The development of end-to-end processes can help reduce waste and increase the value adding activities (Rosemann, 2015). Furthermore, process management can help an organisation achieve its goal and identify the irrelevant goals and processes that do not add value to the customers and organisations at large. In addition, there are operational benefits linked to effective process management such as consistency in high level of operation, speed of operation, time and cost effectiveness as well as delivery of quality product and service which translate into improved customer satisfaction, and organisation's performance (Vom Brocke & Mendling, 2017). Process management also offers various strategic benefits such as effective response to the dynamics of the business environment. Some organisations may not notice decline in performance until it is reflected in the financial performance. Timely recognition of decline in performance may help organisations to implement control measures however, many organisations do not have systems of identifying and responding to changes and decline in performance (Rosemann, 2015).

Krogstie et al. (2005) categorised the benefits of process modelling into three: communication, learning, and long-term advantages (Table 3) while Table 4 presents a comparative analysis among the different process management tools such as BPM, BPR, BPI and TQM.

Other advantages of BPM include (Reijers, 2021; Salvadorinho & Teixeira, 2021; Geyer & Fourie, 2015; Aidara et al., 2024; Morgan, 2020),

1. It promotes a collaborative work environment rather than working in silos.
2. It helps organisations to standardise their operations in line with global best practices.
3. BPM organisation to embrace and adapt to the dynamics of business operations and the changing market environment.

Table 3. Benefits of process modelling (Vom Brocke et al., 2016).

Category	Benefits
Communication	• It fosters team building, effective and healthy relationships, cooperation and motivation among the management and employees. • The model promotes conflict resolution, and better collaborations. • It indicates how communication can be initiated and facilitated.
Learning	• It promotes knowledge acquisition and diffusion about the processes. • It uncovers new information that improves individual's and organisations performance • It promotes human capacity development • Training via process models is time and cost effective
Long-term benefits	• Simple and effective diagrams simplify processes and identify those activities that are important for an organisation to achieve set goals • The models help in sorting people's competence in an organisation. • It aids knowledge acquisition and diffusion. • Models are useful in marketing and in projecting future performance of an organisation. • It promotes effective monitoring of processes and control

Table 4. Comparison among BPM, BPR, BPI and TQM (Reijers, 2021).

BPM	BPR	BPI	TQM
High process efficiency	Short processing times	Reduction in processing times	Reduction in process cycle time
Reduction in turn-around time	Improvement in performance and productivity	Improvement in performance and productivity	Increased flexibility
Cost effectiveness	Cost effectiveness	Cost effectiveness	Reduction in cost
Improved transparency	End-to-end process re-design	Rapid responsiveness	Rapid responsiveness
Improved team and customer	Improved customer satisfaction	Higher customer satisfaction	Improved customer satisfaction
Increase in value adding activities	Removal of process waste	Increase in value adding activities	Removal of process waste

4. It promotes the culture of continual monitoring of activities and effective response to customer requirements or deviation from the standards or targeted goals.
5. It promotes proper alignment with organisation's goals, and also to establish the inter-dependencies among them.
6. It promotes an atmosphere of collaborative engagements and provides organisations with a common language.

It simplifies operational intricacies and provide with a clear workflow as well as useful insights on how business activities are conducted. Dumas et al., (2018) illustrate the process architecture of the British Telecom in different levels. These include: business levels, process levels and operational level.

Business processes can be divided into various sub-processes according to the various levels. The process architecture offers a structured way of organizing and aligning organisation's activities by dividing the processes into three main hierarchical levels namely business, process, and operational as shown in Figure 4. The business level (first level) comprises of the organisation's strategy, process, and performance measurement. It defines the organisation's mission, core values and propositions. This level describes organisation's processes in details highlighting how they create values for the customers and stakeholders. The process level (second level) consists of process design or redesign, as well as implementation of continuous improvement such as lean or six sigma. At this level, the core values defined at the business level are translated into end-to-end business processes are identified major activities, roles, and interactions across the various functions and departments. It focuses on coordination as well as how processes are analysed and performances measured. The operational level (third level), comprises of technology and resource development, monitoring and human capacity development amongst others (Figure 4). The operational level is the implementation phase which deals with the management of business activities (Quintin, 2021; International Organisation for Standardization \ISO 9001:2015\, 2019; Malinova et al., 2014). At the operational level, tasks are specified, decisions rules are made, inputs and outputs are assigned and the system is made to effectively interact. This stage supports the execution of processes including process automation, training, and compliance. The integration of these three levels ensures strategic alignment enabling effective management of business processes, performance measurement and continuous improvement.

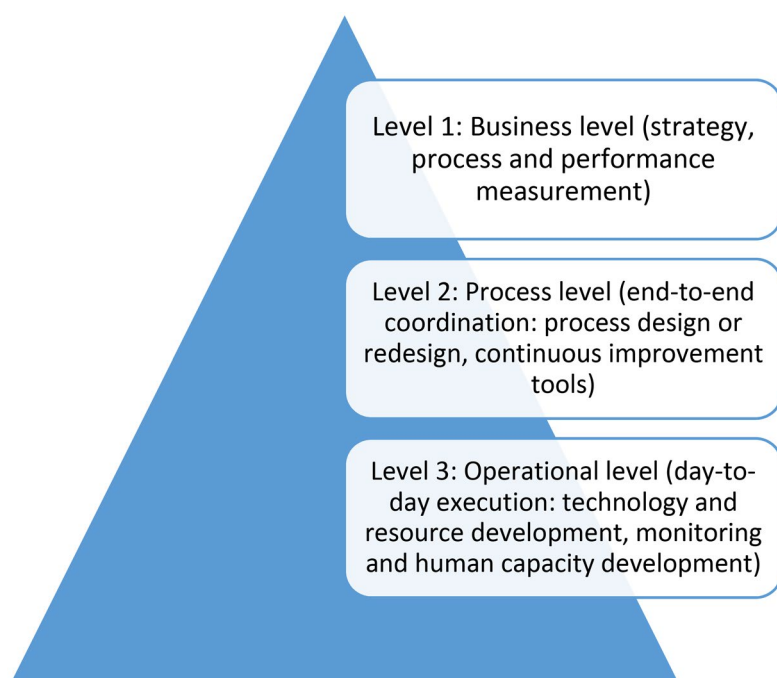


Figure 4. Processes architecture (Dumas et al., 2018; Reijers, 2021).

The Supply Chain Operations Reference (SCOR) framework, is an established business process framework for supply-chain management (Xu et al., 2008). SCOR establishes a standard diagnostic tool as well as supply-chain architecture and establishes metrics for each of the SCOR framework's processes (Millet et al., 2009). SCOR as a management tool span from an organisation's suppliers to ensure that the customers need of customers are met at all stages of the process. This reference model assesses the business processes and activities that occur across the supply chain. The three core pillars of the SCOR framework include: process modelling, performance measurement, and implementation of best practices (Xu et al., 2008). Thus, the SCOR framework allows an organisation to effectively assess the supply-chain architecture to identify the relevant strategies that an organisation can use to drive the set goals (Indulska et al., 2009). Millet et al. (2009) indicated that the SCOR model can be aligned to the business processes and information system, as a result, an organisation must not only develop a process architecture, but also acquire the relevant data to determine the performance and adequacy of the current process (Millet et al., 2009).

The BPM life cycle is a continuous and iterative process that ensures business processes are consistently aligned to the organisation's strategy, and to ensure that the processes respond effectively to changes to deliver sustained value.

The six steps of the BPM lifecycle provided by ABPMP include: planning and development of strategy, process analysis, process design and modelling, process implementation, measurement and control, and process optimisation (Association of Business Process Management Professionals, 2020). These steps are shown in Figure 5. The planning and strategy stage is the initial stage which defines the need for process management and the specific goal the organization intends to achieve. This stage ensures the translation of strategic objectives into process goals, alignment between business strategy with operational activities. It involves the identification and prioritisation of major based on their effect on organisation's performance, customer value, and compliance requirements. The second stage (process analysis) seek to understand the operation of the current process ('as-is' processes). At this stage, the various workflows, roles, resources, information are examined to identified challenges, inherent risk, wastes and inefficiencies to obtain an evidence for initiating process improvement. At the third phase (design and modelling), the improved or redesigned version of the 'as-is' processes is developed called the 'to-be' processes. This involves the development of process models using standardized notations such as the BPMN for the representation of activities, decision points, and interactions with the aim of designing efficient processes that are more customer-focused, compliant and aligned to the organisation's strategic objectives. At the



Figure 5. Steps in BPM life cycle (Association of Business Process Management Professionals, 2020).

process execution stage, the designed to-be' processes are implemented. This stage encompasses workflow deployment, allocation of roles and personnel. Depending on the level of organisation's process automation, processes can be executed manually, semi-automatically, or in the fully autonomous model via the BPM systems and workflow engines. The measurement and control stage employs the key performance indicators (KPIs) such as quality, lead etc. to measure the performance of the improved process to verify if it operates as intended, while control systems are put in place to identify errors and deviations to prompt timely corrective actions. At the process optimisation stage, the performance data and feedback gathered are employed for process continuous improvement. This may include continuous refinements or redesigns to promote process efficiency, agility, or resilience. This stage closes the loop by feeding the insights gathered back into the planning and strategy, thereby reinforcing the continuous improvement process (Association of Business Process Management Professionals, 2020).

Existing studies have highlighted the application of BPM. For instance, Malinova et al. (2014) developed a model for assessing the success of BPM implementation while Volik et al. (2021) developed a methodology for improving organisation's business processes. Ariouat et al. (2017) presented a conceptual model for BPM while Geyer and Fourie (2015) presented a framework for the determination of the appropriateness of a business process modelling technique for supermarket application in South Africa. Loriston et al. (2017) developed conceptual practical guidelines for BPM practice for improving the flow patients in emergency units in the hospitals while Botha and Rensburg (2010) proposed an improvement model with which integrates business process with customer experience management.

The outcome of these studies that BPM is critical to an organisation's performance and suggest practical and conceptualised to aid BPM implementation in organisations.

3.6. Business process management life cycle

The BPM lifecycle summaries the various steps of BPM (Morgan, 2020). This lifecycle will aid the understanding of the role of technology and other enabling factors in BPM implementation and process

management (Dumas et al., 2018). Dumas et al. (2018) indicate that technology is a major factor required for business processes improvement. Technology, principally information communication technology (ICT), is a major driver for improving business operations. It is a useful tool for automating business processes and for controlling and executing business processes to achieve optimum performance. However, technology alone not solve the problems of business operations as there is a need to understand how business processes can be designed and enhanced. This will clarify the roles of technology, in improving the efficiency of business operations.

Figure 6 shows the BPM life cycle designed to provide solutions to process related problems and for process improvement. It focuses on achieving business goals and resolving business challenges. The process architecture starts with process identification and the evaluation of the current (as-is) process while process analysis provides an insight into the weaknesses of the current process and the impact on the overall business. The current process is therefore redesigned to the expected process (to-be process) and executed at the implementation stage. This stage encompasses the organisation's transformation and process automation. This is followed by the process monitoring and control phase to ensure conformity to requirements of the expected process and standards. Following the implementation of the redesigned process, process performance will be measured via the collection and analysis of data to access the performance of the redesigned process vis-à-vis the set objectives.

The BPM lifecycle describes the stages of business processes management and explains how a BPM project can be arranged to improve processes. The lifecycle serves as a guide for business analysts to complete process categorisation into areas, groups, steps, operations, improvements required, and planned future changes (von Bertalanffy, 1968; Szelagowski, 2018; Von Rosing et al., 2021; De Morais et al., 2014; Woliński & Bala, 2018; Cereja et al., 2018).

The BPM lifecycle may not often be implemented applied in strict sequence as its application is a function of the nature of the problem to be solved (Liptak & Liptak, 2003). Thus, any stage that is not relevant to a particular problem to be solved can be excluded. However, the BPM lifecycle is suitable for process analysis or improvement (Woliński & Bala, 2018; Cereja et al., 2018; Matzner et al., 2018) with benefits of precision and accuracy as well as cost and time effectiveness (Dumas et al., 2018).

According to Russack and Menges (2018) process models find application in process analysis, process improvements, certifications system and trainings, etc. Furthermore, Matzner et al. (2018) indicated that the BPM lifecycle find application in problem solving process analysis, redesign, and implementation.

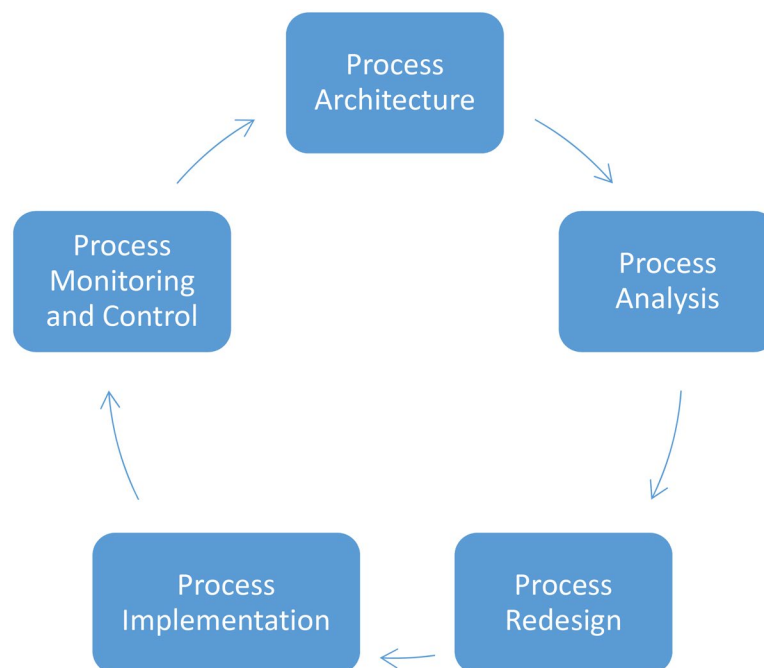


Figure 6. BPM lifecycle (Dumas et al., 2018).

Table 5. BPM lifecycle objectives and methods (Leitz et al., 2018).

Process lifecycle	Objectives	Methods
Process identification	- Define process goals - Evaluate current process - Identify process priority - Identify stakeholders	- Stakeholder analysis - Objective's matrix - SWOT analysis - Interviews - Workshops
Process discovery	- Process documentation - Identify the trends in the current process	- Modelling notation - AS-IF and AS-IS models - Interviews - Workshops
Process analysis	- Identify the limitation of the current process - Identify the process constraints, risks, cost and value	- SWOT analysis - Six Sigma - Lean Six Sigma - value stream mapping - Stakeholder analysis - Activity based and costing - Root cause analysis - Interviews - Workshops
Process re-design	- Define the goal of the new or improved process - Define the requirements of the new or improved process - Identify process capability	- Issues Register - Six Sigma - Interviews - Workshops - TO-BE models - Brainstorming - Reference models - Force Field Analysis
Process implementation	- Embed improved process - Change Management	Project plan
Process monitoring & control	- Process monitoring and control - Process review - Map process capability	- Control chart - Process flow audit data and log files - Service level agreements

Table 6. Application of BPM lifecycle.

S/N	Problem	Method	Outcome	Reference
1.	Poor quality and lack of efficiency in organisation's processes.	Implementation of quality standards for processes using the BPM lifecycle framework as a guide.	Improvement in processes, integration of technology into the organisation's structure and improvement in operational procedure via the use of BPM	Woliński and Bala (2018)
2.	Lack of formal purchasing procedure for medical material at a Brazilian insurance company	Process re-design using the design thinking approach aligned with the stages of the BPM lifecycle framework.	Formalising the employees' insights about the existing purchasing process and the design of a novel (to-be) process for material purchase endorsed by all the stakeholders.	Cereja et al. (2018)
3.	The need to improve patient flow in the emergency department of a hospital in terms of document processing and response time, resource usage, time and cost effectiveness, and patient outcomes.	Process mining aligned with the stages of the BPM lifecycle framework	Development of process models to enable effective flow of patient at the emergency department patient of the hospital. This resulted in significant improvement in patient flow as there exist a little difference between short-stay and long-stay of patients, with timing differences. There was also improvement in terms of document processing and response time, resource usage, time and cost effectiveness. These are major priorities for the delivery of effective health service.	Andrews et al. (2017)

Table 5 presents the objective of each BPM lifecycle phase as well as the applicable methods for implementation while Table 6 presents some of the various application of BPM lifecycle.

From Table 5, it can be seen that the BPM lifecycle model is useful for determining the weaknesses inherent in a process and its impact. Its process analysis and redesign phases, is suitable for process

improvement. It is also compatible with the lean and Six Sigma principles; thus, it enhances organisational performance at a reduced costs and risks.

Hence, the BPM lifecycle is designed for problem solving, by providing process solutions in line with the organisation's objectives including solving business challenges.

However, some of its limitation include the availability of accurate data and information as poor information may lead to poor analysis. If the solution is not properly designed with its methodology implemented according to its guiding principles, it may result in failure.

The outcomes of the implementation of the BPM lifecycle framework highlighted in [Table 6](#) show that is one of the reliable models for process improvement. Due to the fact that the business environment is evolving and dynamic, there cannot be a perfect business process model hence, to achieve to improved performance, it is necessary for organisations to adopt or redesign business process models and tailor it to their specific needs.

Furthermore, the application of the BPM lifecycle framework is a function of the nature of the process to be improved.

The BPM framework describes its initiative according to the goals, process features as well as, organisations and environmental or external features (Liptak & Liptak, 2003). It helps to capture the situations the BPM initiative must align with in a specific context of an organisation.

The four circumstantial dimensions of the BPM initiative are explained as follows (Vom Brocke & Mendling, 2017):

1. Goal perspective – the goal of a BPM initiative influences the planning phase, the corresponding actions to be executed and the strategies be employed.
2. Process features – this must be in alignment with a suitable BPM method or technique. It is based on factors such as such as knowledge, expertise, creativity of the developer as well as the process intricacies amongst others
3. Organisational features – this determines the right approach undertake by the top management to process management. This may take into consideration some factors such as the of the nature of the organisation, size, and culture amongst others
4. Environmental features – this includes a variety of external and environmental factors such as the dynamics of the business environment, the levels of competitiveness or uncertainty amongst others.

The contextual factors are also important for organisational structure, decision making, and behaviour (Vom Brocke et al., 2016). This implies that the BPM framework may likely fail where the contextual factors and situations are not considered.

[Table 7](#) highlights some implementation of BPM framework for business process improvement. BPM framework can be used to enhance its implementation and help to determine if there is a conflicting outcome of increase or decrease in organisational performance. If is flexible and can be adopted to fit different business organisations depending on their peculiarities. It can also provide insights to researchers and practitioners on the BPM initiative to implement. However, it may fail if the BPM methods employed is irrelevant to the context applied. It may also fail if the relevant contextual factors are not identified and incorporated into the BPM framework at an early. It may be challenging for practitioners to identify the relevant contextual factors in their business process modelling initiatives but necessary for the success of the BPM initiative.

3.7. Evolution of the BPM framework

Szelągowski and Berniak-Woźny (2024) identified the major challenges and limitations of the traditional BPM using systematic literature review. The authors found that the current BPM is knowledge intensive, less predictable and less compatible with the evolving industry 5.0 technologies. The authors suggested the need for the assessment and integration of Industry 5.0 requirements including people and organizational culture into the current BPM workflow to promote creativity, innovativeness and value creation.

Table 7. Application of BPM framework.

S/N	Problem	Method	Outcome	Reference
1.	Access to standard and regulation documentation by operational personnel leading to quality issues.	A BPM framework approach was implemented with focus on compliance, improvement, process, organisation, and environment features.	The restructuring of quality system and process modelling letter to improved quality management system. This makes it easy to access the relevant and its regulatory documents.	Krogstie et al. (2018)
2.	Inflexible laboratory processes in line with international standards and the need to digitalise of paper-based work to minimise bureaucratic practices.	The BPM framework introduced provided insights into the relevant contextual factors, such as repetitiveness, interdependence, and variability.	The developed modular architecture permits flexibility	Duelli et al. (2018)
3.	Need for process efficiency to improve customer satisfaction through quick checkout	The BPM framework which analysed the contextual factors and other factors necessary for improving the adopted Radio Frequency Identification (RFID) technology.	The RFID technology was improved resulting in quick check out of customers with improvement in the accuracy of identification.	Leitz et al. (2018)
4.	Need for the improvement in working conditions	A mixed approach comprising of a combination of BPM framework with ergonomic assessment to improve working conditions	BPM can assist in the identification of where interventions are needed to assist in ergonomic evaluations	Pineiro et al. (2023)
5.	Need for continuous improvement in business organisations to enable them achieve quality and bottom-line goal of profitability	Qualitative approach involving purposive sampling	BPM is suitable for continuous improvements in organisation	Tsiri et al. (2022)
6.	Good and precise maps of BPM are missing, thus the need for TomTom like functionality in BPM	Conceptual model of process mining integrated with the The BPM context framework	BPM is suitable for improving business performance	van der Aalst (2009)

Viteri-Sánchez and Novillo-Villegas (2025) propose a framework for sustainable BPM adoption. The proposed framework links the various stages of BPM implementation thus strengthening BPM as a dynamic technique. The proposed framework also emerging technologies from Industry 4.0 and 5.0, thus making it a holistic approach. The findings of Szelągowski and Berniak-Woźny (2024) and Viteri-Sánchez and Novillo-Villegas (2025) are consistent with existing literature. For instance, Koblianska et al. (Koblianska et al., 2023) reported on similar findings that contemporary BPM are now embracing digital tools and technologies citing the role of organisation culture in BPM implementation. Gošnik et al. (2023) added that effective leadership in BPM contributes to organisation's profitability. Kourani et al. (2024) demonstrated the integration of AI into BPM model to improve the interpretability of complex business process while Comuzzi (2024) stated that with the emergence of the process mining, BPM has transitioned from a descriptive model to a predictive and prescriptive data-driven analysis model. Lê (2024) developed an interactive real time dashboard called the 'P-R-dash' as a digital twin component for visualizing and interpreting business process.

Fornari et al. (2025) stressed that digital twin as a digital replica of the physical system can enable a real time interaction, simulation and monitoring of business processes thereby enhancing effective process efficiency and optimization. The authors identify some potential barrier to the digital twin integration as poor scalability and interoperability, limited real-time data integration, and the inadequate alignment of digital models with operational realities suggesting the need for standardized approach to the deployment of emerging technologies such as digital twin, IoT, and simulation techniques in BPM. The work of Fornari et al. (2025) links to that of Garcia et al. (2019) on process mining, which suggests data exploration and analysis in business processes based on real time implementation event logs and deployment of analytical algorithms rather than relying on prescriptive models. According to the authors, the process mining comprises of three major stages namely process discovery, checking for conformance and enhancement. This study corroborates existing studies such as Dumas et al. (2018) BPM can provide details of process behaviour and identify areas of deviation or inefficiencies.

While Garcia et al. (2019) emphasise the use of process mining for process pattern recognition and conformance verification, role of event logs and analytical algorithms for understanding real processes

(discovery), the use of digital twins extends this application by deploying real time data for process simulation. While process mining depends on historical or logged dataset to obtain analytical insights, digital twin utilizes real time data for observing process behaviour (what is happening) and for predictive analytics (what could happen).

Thus, the combination of these tools can strengthen BPM capabilities for diagnostic and prognostics activities as well as continuous monitoring, and adaptive decision-making.

Another emerging technology is the IoT that permits data transmission and sharing in real time. Compagnucci et al. (2023) connect the BPM with IoT by drawing insights from the literature focusing on three key dimensions. The first is the IoT-aware modeling that deals with the frameworks that integrate IoT devices such as sensors, actuators, edge computing into process modeling languages. The second is the requirements for IoT-integrated BPM focusing on the functional and non-functional requirements that can enable business processes interact effectively with real-world IoT data. Lastly, the notation and extensions that describes how existing BPM notations such as BPMN can be adapted or extended to support IoT capability and real-time business process (Compagnucci et al., 2023). Some of the challenges identified by Compagnucci et al. (2023) include lack of standardized modeling languages that is compatible with BPM and IoT intricacies, the challenge of integrating real-time constraints and continuous event streams into the discrete process models, and lack of supporting tool for the simulation and implementation of IoT-aware processes. The work of Compagnucci et al. (2023) advance the broader BPM discourse by highlighting how IoT can transform BPM. This is necessary because the traditional BPM models are static, and abstract, representations of organizational processes while the proposed IoT-aware BPM, models can capture the dynamic interactions with physical systems.

Weske (2024) provides a detailed theoretical explanation on the recent advances in BPM digitalization to achieve a data-driven process management, and distributed systems. The work emphasises the integration of process modeling languages such as the BPMN with implementation architectures or workflow engines. Weske (2024) can form a theoretical basis for BPM, while the proposed process mining, IoT-aware modeling, and digital twins can form the gradual extensions to promote BPM's performance, responsiveness, intelligence, and adaptability in business environments. From the theoretical perspective Weske (2024) emphasises that BPM can enable enables organizational efficiency and transparency, regulatory compliance, and agility, especial in a complex and dynamic business environment. All these contemporary studies describe the evolution of BPM from a static, model dependent into a dynamic, data-driven, and cyber-physical process intelligent technique.

4. Application of business process management for risk mitigation

The business environment is a highly competitive environment where business process modeling, analysis and re-engineering is necessary because of the increasing risks, dynamics of the business environments, customer requirements, and stringent nature of policies, regulations and standards. Nikolaidou et al. (2002) conducted the modelling and automation of the business processes geared towards loan monitoring in a medium-sized bank. The BPM was achieved using the Modified Petri-Net model. The outcome of the study indicated that the BPM is a feasible approach for monitoring loan services in the banking industry.

The business operations require delivery of quality product and services with speed and agility hence the need for BPM. The effective deployment of BPM can facilitate significant performance improvement in business performance in terms of operations and fraud mitigation amongst others. The dynamic business nature, makes risk inevitable. However, the implementation of the BPM can reduce the vulnerabilities of organisations to the business risks.

This study proposes a BPM framework comprising of the basic components that organisations can use to identify, assess, and manage risks proactively. The process mapping and modelling, can be used to identify risks so that preventive measures can be taken. Automated workflows can be used to reduce the risk of errors and irregularities, while real-time monitoring and alerts can help in the quick identification of potential threats, so that they can be mitigated. Furthermore, the proposed BPM framework can be used to provide detailed documentation and investigation trails, which forms an important aspect of the risk management and legal framework. It is holistic comprising of the various phases of risk management (Figure 7).

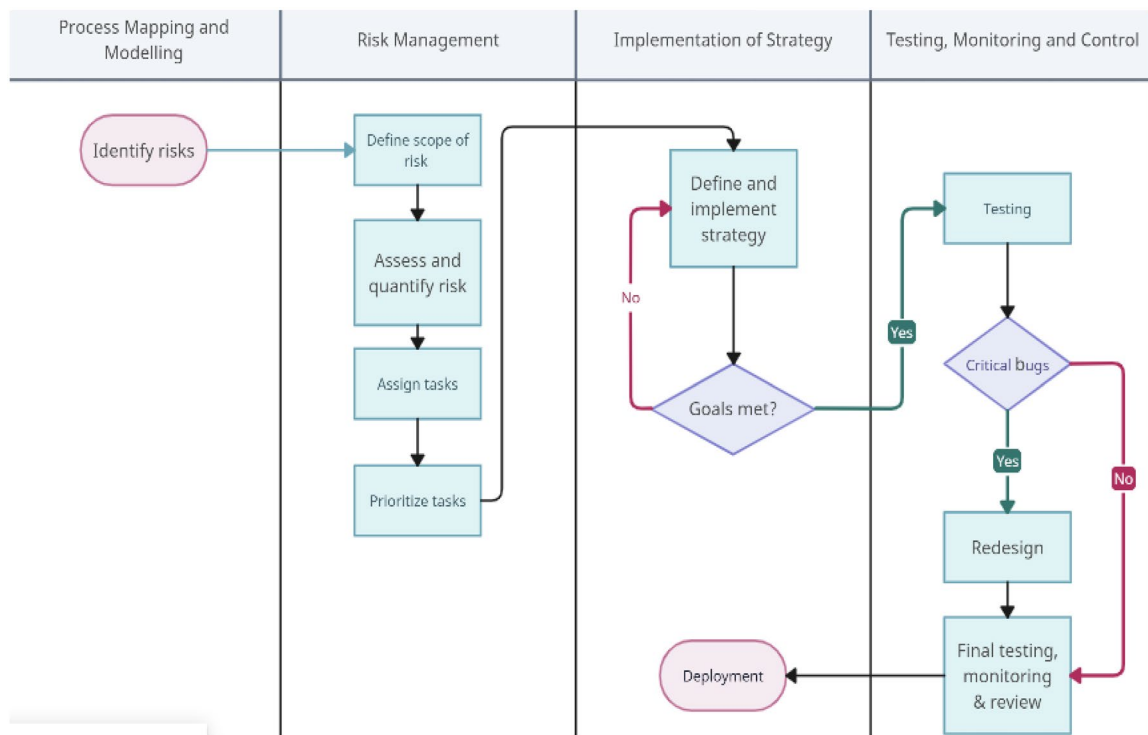


Figure 7. Proposed BPM framework for risk mitigation.

Source: Author's.

Table 8. Specific BPM tools or techniques that can be applied within each step of the risk mitigation framework (Leitz et al., 2018; Chang et al., 2023).

Framework step	BPM tools/techniques applied	Purpose
Risks identification Definition of scope of risk	BPMN, process maps Process mining, BPMN swimlanes	To visualise and locate the exact risk points To identify the boundaries of risk, and empirical scope
Assessment and quantification of risk Assignment of tasks	Simulation, process mining BPMN swimlanes, RACI (Responsible, Accountable, Consulted, and Informed) tool	To quantify the impact of the risk impact Useful for accountability and governance
Prioritisation of tasks	Simulation, process mining	Useful for ranking risks by severity or likelihood
Definition and implementation of strategy	BPMN, EPC (Engineering, Procurement and Construction), simulation	For documentation and validation of the strategies employed
Testing (monitoring and control)	Process mining, simulation	For real-time monitoring and stress testing
Redesign and final review	BPMN, mining, simulation,	For iterative improvement
Deployment	Process maps, mining	For ensuring standardization and compliance tracking

Table 8 presents the details of how specific BPM tools or techniques, that can be applied within each step of the risk mitigation framework shown in Figure 7.

The proposed framework in Figure 7 integrates risk identification, assessment, strategy implementation, and monitoring within a BPM perspective. Its effectiveness can be explained by aligning it with the six core elements of BPM as defined by Rosemann and Vom Brocke (2015), as well as contextual factors and lifecycle considerations.

4.1. Strategic alignment

It starts with the identification of risks and definition and implementation of strategy. This ensures that risk management activities are directly linked to specific organisational goals. Strategic alignment ensures that processes support the organisation core objectives (Dumas et al., 2018). By embedding risk assessment into the process mapping, the organisation can align operational resilience with their broader strategic priorities.

4.2. Governance

The steps that deals with assigning and prioritising of tasks ensures accountability and effective decision rights in risk management. BPM governance offers structures for process ownership and performance monitoring (Dumas et al., 2018) and with clearly defined roles or responsibilities, the proposed framework will ensure compliance to standards and reduction in ambiguity while handling risks.

4.3. Methods

The flow of activities from risk assessment to the testing and redesign phase show a form of structured method for risk mitigation. BPM underscores methodological rigour via modelling techniques, key performance indicators, and continuous improvement (Vom Brocke et al., 2016; Vom Brocke & Mendling, 2017). Hence, the decision gateways (such as 'Goals met?'; 'Critical bugs?') exhibit the application of analytical methods to support an iterative decision-making.

4.4. Information technology

The processes of testing, monitoring, and control are strongly enabled by information technology. Modern BPM relies on digital information systems, simulation, and automation tools (Cardoso & Aalst, 2009; van der Aalst, 2009). Thus, in this proposed framework, IT underpins the processes of risk assessment and quantification, testing, and redesign, to ensure a data-driven monitoring and rapid detection of risks.

4.5. People and culture

There is a need for skilled expertise or personnel to effectively execute the processes of risk mitigation and maintain a culture of collaboration. For instance, the assignment and prioritization of task reflect human roles in risk management. Literature on BPM literature highlights the roles of people as both executors and innovators (Cardoso & Aalst, 2009; Dumas et al., 2018; Delavari et al., 2010; van der Aalst, 2009). Moreover, the testing and redesign phases are looped signaling a culture of continuous feedback and improvement.

4.6. Contextual factors

Context influences the rigour applied in the implementation of the proposed framework. Factors such as industry's regulations, organisation's maturity, and external pressures can determine the BPM adoption (Cardoso & Aalst, 2009; Dumas et al., 2018). Depending on the perceived context by adopting organisation, priority can be set. For instance, financial institutions may prioritise security, compliance and governance, while information technology firms prioritize may prioritise innovation and agility. However, the adaptability of the proposed to different organisation context demonstrates its contextual relevance.

The proposed framework aligns with BPM lifecycle stages shown in Figure 6 (Dumas et al., 2018). For instance, the process architecture, analysis, design and modeling (encompassing activities such as *risk* identification, scope definition as well as assessment and quantification of risk), while the implementation phase assigns and prioritise tasks and implement specific strategy. This is followed by the monitoring and control phases which tests, and debug leading to optimisation and deployment. This lifecycle perspective highlights the iterative nature of the proposed BPM framework where risk management is embedded as a continuous cycle of improvement.

The proposed BPM risk mitigation framework offers several novel contributions compared to the existing framework in BPM. For instance, it integrates risk management with the BPM lifecycle by embedding risk identification, quantification and mitigation thus introducing a more proactive approach to each stage of the design and implement. This is an advancement over the traditional based BPM models which emphasise design, execution, and optimization (Dumas et al., 2018) only. Furthermore, the proposed framework aligns with the six core elements of the BPM unlike the conventional BPM risk tools

that focus on information technology or governance or aspects only. This framework explicitly connects risk activities to all the six BPM core elements. This ensures a holistic detailed and multi-dimensional approach to risk management. It is decision oriented via iterative feedback loop unlike the traditional frameworks and emphasises contextual factors by indicating risk prioritization, and governance structures, which can be adapted to various domain-specific needs. This context makes the proposed framework useful across various industries with varying regulatory and operational environments.

5. Results and discussion

This section discusses the key findings drawn from the systematic review and provides answers to the research questions.

5.1. Summary of findings

The results obtained indicate that BPM is a renowned technique employed by organisations to improve business process, and efficiency and also to promote transparency, and strategic alignment among the various units and actors. It also finds a wide range of application across various organisations.

The outcome of the review indicates that BPM frameworks utilize a lifecycle-based approach, comprising of the planning, analysis, design, implementation, monitoring, and optimization phases. This enables organisations to effectively manage end-to-end processes and to implement continuous improvement processes (Dumas et al., 2018; Weske, 2024). The findings from different studies show that organisations deploying BPM experience improvements in process visibility, reduction in process variability, lead time reduction, and improved regulatory compliance.

Furthermore, the outcome of the study reveals that there is a gradual and systematic shift toward data-driven BPM. This is evidenced in the increasing rate of implementation of process mining and simulation techniques to determine actual process performance or behaviour over time to identify areas of deviations. These techniques enable evidence-based decision-making (Chang et al., 2023). Recently, studies have reported on the deployment of emerging technologies in BPM. These include: Internet-of-Things-enabled processes, real-time analytics, and digital twins. This enables BPM as a predictive and adaptive process management (Fornari et al., 2025; Hammer, 2010).

In addition, the findings show that the traditional BPM frameworks are mostly based on modelling approach and are technologically driven. This may not sufficiently capture the social, organizational, and human factors in business processes and management, thus, making organisations resistant to changes and sometimes resulting in reduction in user adoption, or reduction in sustainability improvements (Trist & Bamforth, 1951; Bostrom & Heinen, 1977).

To overcome this challenge, the Dalberg process modelling framework was introduced capturing both formal workflows and informal practices (APQC, 2020). Although it is less formal compared to the traditional BPM but it can complement it by capturing real-world business scenarios rather than the ideal process flows.

The results indicates that BPM is a tool for process documentation, automation, as well as strategic and socio-technical management tool. Theoretically, the findings align with the system's theory, that views an organisations as an interconnected process systems that requires effective coordination for optimal performance (von Bertalanffy, 1968). The end-to-end process architectures in BPM practice is a strong reflection of its alignment to the system's theory. The results also indicate that the BPM outcomes are explained by the organisation's theory. Organisation's structures such as governance, organizational culture, and stakeholders and institutional contexts can significantly influence the outcome or success of BPM (Scott & Davis, 2000). Hence, BPM frameworks must be deployed based on the various organisation's context rather than being deployed uniformly (Donaldson, 2021). The success of BPM is also a function of social and technical subsystems as explained by the socio-technical systems theory. This theory advocates for consideration of human factors such as skills and user participation etc. as a significant contributor to the long-term success of BPM (Trist & Bamforth, 1951; Bostrom & Heinen, 1977).

Recent years have witnessed the transformation of BPM tool into an adaptive, data driven process management tool with the integration of emerging digital technology such as the digital twins into it. This marked a major evolution in BPM and extends its range of application. to simulation and predictive analytics with the integration of real-time data. This makes the technique more responsive and adaptive to organisations need although it raises ethical concerns and data quality issues. On the overall, the findings indicates that the future of BPM frameworks will move from a static process modelling tool to a more sophisticated data-driven and adaptive tool integrating emerging technologies such as artificial intelligence (AI), process mining, Internet of Things (IoT), and digital twins. This will make it a more robust tool for long-term sustainability and deployment in an increasingly complex business environment.

Table 9 presents the summary of the findings on BPM according to the major themes.

5.2. Answers to research questions

This section provides answers to three research questions underlying this study are:

1. RQ1: What are the existing theories, frameworks and approaches used in BPM?

Evidence from the literature synthesized indicates that BPM is rooted in multiple theoretical frameworks such as the system's theory, organisation's theory and socio-technical system's theory. The systems theory provides the structural and integrative basis of BPM while the organization's theory highlights the influence of organisation's factors such as context, governance, and culture on BPM outcomes. Lastly, the socio-technical systems theory advocates for the inclusion of human factor into the BPM framework to ensure that the ensures BPM technique is are technically sound, sustainable and socially acceptable.

The integration of these theories provides a theoretical basis for the design and implementation of BPM in an organisation.

Table 10 summarises the answer to the first research question.

In terms of frameworks employed in BPM, studies indicates that BPM employs structured frameworks such as BPM Lifecycle Framework comprising of six phases namely planning, analysis, design, execution,

Table 9. Summary of the findings on BPM.

Theme	Key findings	Reference
BPM significance	For business process modelling to improve efficiency and also to promote transparency, and strategic alignment. It also improves process visibility and compliance. The end-to-end capability promotes continuous improvement.	Dumas et al. (2018); (Garcia et al. (2019); Weske (2024)
BPM Lifecycle	BPM lifecycles consists of phases such as planning, analysis, design, execution, monitoring, optimization.	Weske (2024) Dumas et al. (2018)
Theoretical frameworks	BPM is supported by the system's theory, organisation's theory and socio-technical systems theory	Scott and Davis (2000) Trist and Bamforth (1951); Bostrom and Heinen (1977)
Trend: Integration with emerging technologies	BPM is compatible with emerging technologies such as digital twins and process mining which transforms it an adaptive and data driven technique. This enables it to identify process behaviour and deviations, enables predictive and analytics and identifies deviations, and supports evidence-based decision-making.	Garcia et al. (2019); Fornari et al. (2025)
Limitation of BPM	Traditional BPM is limited by inadequate real-time responsiveness, limited consideration for human and organisation factors, poor scalability, poor standardisation and integration with IoT and cyber-physical systems, and limited adaptiveness	Dumas et al. (2018) Fornari et al. (2025) Compagnucci et al. (2023)
Future: Integration with more emerging technologies	BPM may witness integration with more emerging technologies such as AI, IoT, and digital twins to improve its responsiveness and adaptivity	Fornari et al., (2025); Compagnucci et al. (2023)

Table 10. Theories underlying BPM implementation.

Theory	focus	assumption	Strength	Limitation	Application in BPM	Reference
System's Theory	Emphasises organization as an open system comprising of interrelated and interdependent subsystems	Processes are interconnected and overall system performance is a function of the feedback received and environmental interaction	Promotes an integrated end-to-end process	Limited consideration for human and institutional factors	Useful in process architecture, end-to-end BPM design, design of KPI feedback loops, and in continuous improvement	von Bertalanffy (1968); Weske (2024)
Organisation's Theory	Emphasises organisational factors such as structure, governance, culture, as well as stakeholders and institutional context	Process performance is a function of effectiveness depends on organisational context. No single best fit process for all organisations	Promotes effective coordination and accountability	Limited consideration for human factors	Useful in process governance , ownership, change management, and ensures flexibility for a customised context-specific BPM implementation	Weske (2024) Donaldson (2021) Scott and Davis (2000)
Socio-Technical Systems (STS) Theory	Emphasises integration of social and technical subsystems	Organisation's performance is a function of strategic alignment between people, processes, and technology	Promotes human-centred and sustainable BPM implementation. It reduces resistance to change and improves adoption	Less formal and complex to operate	Useful in process automation, integrates digital transformation tools such as process mining, IoT and digital twins	Trist & Bamforth, (1951) Bostrom and Heinen (1977); Fornari et al. (2025)

monitoring and optimization (Dumas et al., 2018; Weske, 2024). Furthermore, BPM uses process architecture frameworks to organise business processes across the business, process, and operational levels (Dumas et al., 2018; Reijers, 2021). It also uses the Porter's Value Chain to link the business processes to value creation to gain a competitive advantage (Dumas et al., 2018) as well as the Dalberg Process Modelling Framework for end-to-end and stakeholder-driven process analysis (APQC, 2020). Recent studies such as Fornari et al. (Fornari et al., 2025) have reported on emerging digital BPM framework driven by data and digital twin to extend BPM application to real-time monitoring, predictive analytics and simulation.

In terms of approaches used in BPM, studies reported on the use of process modeling, BPMN, and activity diagrams for formal process representation (Dumas et al., 2018). Furthermore, the Lean Management techniques are used for waste identification and elimination while the Six Sigma –is used to mitigate process variability and to improve product quality. The TQM is also used for continuous quality improvement (Dumas et al., 2018). In addition, process mining technique is used to discover patterns, trends and to check for process conformity and status over time (Ivančić, 2019; Andrews et al., 2017; Koblianska et al., 2023).

2. RQ2: What are the trends, benefits, and limitations of BPM?

In answering the second research question, BPM has evolved from a static business process model into a data-driven and digitally enabled business management technique with the integration of emerging technologies such as process mining, automation and digital twin. These technologies enable it to perform real-time monitoring, and predictive analytics. Literature studies found that it delivers significant benefits, such as improved quality and operational efficiency, increased process transparency, and optimisation leading to increased customer satisfaction. Furthermore, BPM supports continuous improvement, fostering better alignment of processes with strategic objectives. Despite these benefits, the implementation of BPM still faces some challenges across various organisations. For instance, depending on organisation's context, it could be resource-intensive, complex to implement, and may lack the required flexibility and adaptability thus leading to resistance to change especially the traditional BPM that lacks consideration for human factors, leading to resistance to change. In additional, the effectiveness of BPM effectiveness is also a

function of the organization's factors such as organisation's culture, and governance, and resources. Some of these trends, benefits and limitation of BPM are summarized in Table 11.

3. RQ3: What are the potential applications of BPM and the gaps existing in its framework?

BPM offers significant applications across various organisations such as business, manufacturing, banking and finance, public administration etc in the areas of process design, implementation, monitoring and improvement thus, contributing to in process automation, compliance, and product quality improvement. Although the traditional BPM framework is still limited because it is not fully digitally inclined and data-driven. However, digital applications such as process mining, real-time analytics, digital twin and IoT-enabled processes have evolved for predictive and adaptive process management. Despite the benefits and applications of BPM, there still exist some gaps such as limited real-time responsiveness especially in the evolving world that is digitally inclined and data-driven. Traditional BPM frameworks are limited in scalability and deployment in IoT-enabled and cyber-physical systems. There is also insufficient consideration for human and organisation's factors. These gaps necessitate the need for a more adaptive, socio-technical, and data-driven BPM frameworks.

Table 12 summarises the potential applications of BPM and the gaps in the existing frameworks.

Table 11. Some of these trends, benefits and limitation of BPM.

Dimension	Description	Literature
Trends	Increasing adoption and implementation of BPM across various sector.	Weske (2024)
	Increased integration of BPM with emerging technologies such as process mining, automation, AI, IoT and digital twins.	Fornari et al. (2025) Garcia et al. (2019)
	Increased shift toward a data-driven BPM for real-time monitoring, and predictive	Compagnucci et al. (2023)
Benefits	Increasing emphasis on end-to-end processes and human-centered design	
	Improvement in operational efficiency	Dumas et al. (2018)
	Increased process transparency, and optimisation	Weske (2024)
	Increased process compliance and enhanced product quality.	Hammer (2010)
	Increase in customer satisfaction	
	Improved alignment between strategy and organisation	
Limitations	Improved support for continuous improvement and organization's resilience	
	Improved agility, flexibility and adaptability	
	High cost of BPM implementation and maintenance	Weske (2024)
	Limited human factor	Scott and Davis (2000)
	Organisational resistance to change	Trist and Bamforth (1951)
	Dependence on organisation's factor	Bostrom and Heinen (1977)
	Challenges in integrating human and technical factors	

Table 12. Potential applications of BPM and the gaps in the existing frameworks.

Application	Description	Literature
Process design	Modeling, analysis, and end-to-end business processes design	Dumas et al. (2018) Weske (2024)
Process automation and implementation	Workflow automation, Integration of information system and other sub-systems	Dumas et al. (2018) Weske (2024) Hammer (2010)
Performance and regulatory compliance	KPIs monitoring KPIs, regulatory compliance, auditing, and process monitoring and control	Dumas et al. (2018) Scott and Davis (2000)
Pattern recognition, predictive analytics and simulation	Adoption of emerging digital technologies such as process mining, AI, IoT-real-time analytics, and digital twins for adaptive and predictive analytics, modelling and simulation	Weske (2024) Fornari et al. (2025) Garcia et al. (2019) Compagnucci et al. (2023)
Customer experience and service delivery	Quality service delivery Effective responsiveness, and consistency in business processes	Weske (2024) Hammer (2010)
Gaps	Description	Literature
Human and Social Factors	Limited integration of human and organisation's factors such as organisation's culture, skills, learning, user adoption into the BPM frameworks	Trist and Bamforth (1951) Bostrom and Heinen (1977)
Real-Time and Unstructured Processes	Limited support for real-time decision-making Knowledge-intensive High process variation	Weske (2024) Fornari et al. (2025) Garcia et al. (2019)
IoT and Cyber-Physical Integration	Lack of standardized notations and architectures for emerging data-driven frameworks such as IoT-enabled and cyber-physical processes	Compagnucci et al. (2023)
Sustainability and Resilience	Limited consideration for environmental, social, and resilience metrics in the BPM models	Dumas et al. (2018) Fornari et al. (2025)

5.3. Emerging and future trends

The proposed BPM risk mitigation framework is positioned for future developments because it is technology-enabled having integrated modelling and simulation. It is agile and iterative with decision gateway and human centered with consideration for culture, people, and governance. In addition, it is context-sensitive which makes it adaptable across different industries and regulatory environments. These features make it scalable, resilient and adaptive to risk management.

Modern BPM frameworks are increasingly driven by digital technologies and enabled by a culture of data management for effective decision-making in real time. The integration of smart sensors, AI or ML models will make the BPM framework intelligent (Intelligent BPM) and the application of robotic process automation (RPA) for predictive and adaptive process management will make the process accurate and time effective. Through process mining and analytics, events logs can be used to provide real-time insights for process implementation. Furthermore, emphasis in recent times have been laid on flexibility and iterations to enable processes adapt quickly volatile business environments and become resilient to disruptions. In addition, BPM frameworks could also integrate environmental, social, and governance (ESG) metrics into process design and evaluation to promote sustainability. Hence, the next few years may witness a more robust, intelligent, agile, adaptable, sustainable and data-driven BPM frameworks.

Since data is central to business processes, the future trends of BPM and the enabling framework may be data-driven featuring increased AI and ML for predictive analytics and decision making as this is central to BPM (Matzner et al., 2018). It could also feature real-time process monitoring and control of processes through process mining and Digital Twins (DT). Organisations are gradually embracing agile systems and processes that have the capacity to continuously adapt to changing business environments (von Bertalanffy, 1968) as well as systems that facilitates rapid design to ensure resilience to disruptions, hence, the future of BPM may also witness more resilient, flexible and adaptable systems and processes that can respond effectively and in real time to the dynamic nature of business processes and environments. Rosemann and vom Brocke (Rosemann & Vom Brocke, 2015) indicated that the BPM nature is human-centric integrating stakeholders, customer experience, and organisation's culture into process design. The future of BPM could be more collaborative and could also witness increasing human-robot or human-machine collaborations. For effective storage of information and transmission in real time, the future of BPM could witness the integration of technologies such as cloud computing for cloud storage, digital twins, AI, and Internet of Things (IoT) for effective simulation, data analysis and sharing in real time (Fornari et al., 2025; Koblianska et al., 2023; Kourani et al., 2024; Comuzzi, 2024; Garcia et al., 2019). These technologies will improve the process of data management ensuring transparency, data integrity and fast decision making. On the overall, the findings indicates that the future of BPM frameworks will move from a static process modelling tool to a more sophisticated data-driven and adaptive tool integrating emerging technologies. This will make it a more robust tool for long-term sustainability and deployment in an increasingly complex business environment.

Thus, the future trend of BPM may be digitalised featuring the integration of more enabling digital technologies such as AI, digital twins, robotics, cloud computing and IoT thereby making the BPM processes more resilient, flexible and adaptable.

6. Conclusion

This study explores the potentials and benefits of business process modelling, via a literature survey. The literature review conducted highlighted the usefulness of the business-modelling framework, its notation, outcome of its implementation by various organisations as well as practical guidelines for its implementation.

Initially, a total number of 5630 articles were obtained from different academic database using different search engines and only 60 articles were reviewed due to their relevance to the topic discussed. The review highlights the importance of the business-modelling framework, as well as its notation, and some outcomes from various organisations where it has been implemented. The findings from the literature shows that BPM is crucial to the sustainability, monitoring and evaluation, standardisation of

business amongst other benefits. It can assist organisations to understand improved ways to optimise workflows, analyse and automate current business process. Based on the outcomes of the literature review, this study proposes a BPM framework comprising of the basic components that organisations can employ to identify, assess, and manage risks proactively. The outcome of this study can assist in the visualisation and documentation of workflows, process automation, identification of key performance indicators, identification of success and failure rates of business processes, identification of the timeline of each business process and the overall process, identification of potential challenges inherent in business processes and ways to resolve them. This can further aid an organisation's decision-making processes.

6.1. Contribution to knowledge

This study contributes to the existing studies on BPM by identifying the existing theories, supporting BOM implementation as well as the commonly used frameworks and approaches used in BPM? It also delves into the trends, benefits, limitations, and potential applications of as well as and the gaps existing in its framework. It contributes a conceptual and a practical guideline for BPM implementation for business risk mitigation. It advances BPM theory by linking the system's theory, organisation's and socio-technical, system, theories into the traditional BPM framework. It also advocates for the incorporating of emerging digital technologies into the BPM framework thus transforming BPM into a data-driven, adaptive, and human-centered management framework. Unlike previous studies which considers BPM from a theoretical perspective, this study highlights the application of integrated theoretical framework in a digitally enabled and dynamic business organizational environments.

Additionally, this study offers some practical policy recommendations for the deployment of digitally enabled data -driven BPM grounded in integrated theoretical framework. These recommendations can assist top management, process designers as well as decision and policy in the implementation of BPM that is adaptive, efficient, human-centered, data-driven and sustainable.

6.2. Policy recommendations

Drawing from the outcome of the synthesized literature the following are some policy recommendations that can assist organisations to effectively deploy BPM:

6.2.1. Adopt of data-driven BPM tools

In this digital era, where business and organisations are driven by data, the traditional BPM upgraded by integrating emerging digital technologies such as process mining, AI, IoT and digital twin to enhance efficiency and promote a data-driven process. To achieve this, there is a need to build human capacity and expertise through employee training in these areas because they are scarce skills. This will imbibe a culture of process analytics, AI or IoT-enabled process modeling, and digital twin simulation the organisation.

6.2.2. Develop standards and guidelines for contemporary BPM

Contemporary BPM lacks standards and guideline for implementation thus limiting its deployment. The development or adoption of national or industry-level standards for contemporary BPM that integrates AI, IoT and digital twin for process modelling, simulation and real-time monitoring is recommended. This will ensure a seamless implementation, interoperability and foster data security.

6.2.3. Alignment with environmental, social and resilience factors

The traditional BPM has limited consideration for environmental, social, and resilience metrics. It is therefore recommended that organisations integrate environmental, social and resilience KPIs into BPM initiatives for effective performance monitoring and measurement, particularly for sectors like energy, logistics, and manufacturing.

6.3. Limitation and direction for future studies

This study is limited to literature review and the development of a proposed BPM framework for business risk mitigation. Future studies can consider the validation of the developed model within the context of a business organisation. Literature is limited in the area of integration of emerging technologies into the traditional BPM, thus, future studies can also conduct empirical studies on the impact of emerging technologies such as AI, IoT and digital twin on process efficiency, decision-making, and optimization. Furthermore, future studies can investigate the standardized notations and real-time architectures that are compatible with the AI, IoT or digital twin driven BPM in a complex, or distributed environments. Since the organisation's and socio-technical system theories advocates for the consideration of organisational culture, human factor and change management in the BPM workflow, future studies can investigate the impact of these factors on BPM efficiency.

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Authors contributions

CRediT: **Oluwatoyin Esther Akinbowale**: Conceptualization, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Mulatu Fekadu Zerihun**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing; **Polly Mashigo**: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability and materials statement

The data that support the findings of this study will be made available by the corresponding author upon a reasonable request.

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